Hongliang Li

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Education

- **Fudan University — Research Institute of Intelligent Complex Systems** *Starting Aug 2026*
  Ph.D. Student | Biology Foundation Models & AI-Driven Drug Discovery Shanghai, China
  - Advisor: Prof. Shuangjia Zheng
- **Beijing Institute of Technology** *Sep 2020 – Jun 2023*
  M.S. in Computer Science | NLP & Biological Language Models Beijing, China
  - Advisor: Prof. Bin Liu
- **Beijing Institute of Technology** *Sep 2016 – Jun 2020*
  B.S. in Mechatronic Engineering Beijing, China

Professional Experience

- **AI Department, Lingang Laboratory** *Jul 2025 – Jul 2026*
  Research Engineer Shanghai, China
  - Developed and optimized unified affinity-scoring models spanning small molecules, antibodies, and peptides, with multi-target binding prediction capabilities.
- **Institute of AI Industry Research, Tsinghua University** *Sep 2023 – Apr 2025*
  Research Assistant, ATOM Lab Beijing, China
  - Contributed to a Ministry of Science and Technology initiative on deep-learning-based end-to-end protein structure prediction. Advanced neural-network methods for protein conformation generation and structural validation.

Selected Projects

- **AIRFold — Systematic Protein Structure Prediction** Homepage | GitHub
  - Developed at the **Institute for AI Industry Research (AIR), Tsinghua University**, within the ATOM Lab led by **Prof. Yanyan Lan**; contributed to the National Key R&D initiative on deep-learning-based end-to-end protein structure prediction.
  - Helped build a scalable, systematic prediction platform combining homology mining and MSA generation with AlphaFold2, RoseTTAFold2, OmegaFold, ESMFold, and model-quality estimation in a modular microservice architecture.
  - Advanced protein conformation generation, structural validation, and model ranking. The resulting system achieved top-ranking performance in the continuous automated model evaluation (CAMEO) benchmark.

Publications

Total citations: ***404***

- **Hongliang Li** and Bin Liu. (2023). "BioSeq-Diabolo: Biological Sequence Similarity Analysis Using Diabolo Embeddings." *PLOS Computational Biology*, 19(6), e1011214.
- **Hongliang Li**, Yihe Pang, and Bin Liu. (2022). "BioSeq-BLM: A Comprehensive Platform for Analyzing DNA, RNA, and Protein Sequences Based on Biological Language Models." *Nucleic Acids Research*, 49(22), e129.
- Qilong Wang, Huikun Zeng, Yan Zhu, **Hongliang Li**, et al. (2021). "Dual UMIs and Dual Barcodes With Minimal PCR Amplification Removes Artifacts and Acquires Accurate Antibody Repertoires." *Frontiers in Immunology*, 12, 778298.
- **Hongliang Li**, Hongyu Wang, Dongping Chen, and Zhengshe Kang. (2020). "Revealing the Optical Properties of Polycyclic Aromatic Hydrocarbon Clusters with Surface Formyl Groups." *Proceedings of the Combustion Institute*, 38(1), 1207–1215.
- **Hongliang Li**, Zhengshe Kang, and Dongping Chen. (2019). "HOMO–LUMO Gaps of Polycyclic Aromatic Hydrocarbon Clusters with Formyl Groups." *12th Asia-Pacific Conference on Combustion (ASPACC 2019)*.

Research Expertise

- **Biological Foundation Models:** multimodal biomolecular representation learning, protein structure and conformational-ensemble generation, and structure–function modeling.
- **AI-Driven Drug Discovery:** unified affinity prediction across drug modalities, molecular design, hit-to-lead optimization, and multi-target modeling.
- **Scientific AI Systems:** scalable model orchestration, agent-based scientific workflows, provenance tracking, computational validation, and reproducible decision support.